DO NOW — RECALL

# Lesson 129 · Dictionary Compression and Repeated Subs

Name: _____ Class: _____ Date: _____

**1. RLE is effective on what kind of input?**

**2. How many bits are in three bytes?**

**3. Which memory is typically volatile: RAM or ROM?**

**4. What is the binary result of 1 + 1 in a full adder with carry-in 0?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Long consecutive runs of identical symbols.

### 2. Model answer

24.

### 3. Model answer

RAM.

### 4. Model answer

Sum 0, carry-out 1.

### By the end, you can…

• Locate useful repeated words or substrings.

• Assign tokens to recurring sequences and store mapping.

• Replace sequences with tokens; decode using the same mapping.

### Vocabulary — know these terms

dictionary coding — Encodes recurring sequences by referring to dictionary entries.

token — Short code representing a longer sequence.

overlap — Repeated sequence used by more than one part of the source.

lossless — Reconstruction is exact when dictionary and token mapping are known.

NEW LEARNING

# Read, annotate and explain

### Find recurring text

Locate useful repeated words or substrings.

### Build dictionary

Assign tokens to recurring sequences and store mapping.

### Encode and decode

Replace sequences with tokens; decode using the same mapping.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Use dictionary {#1:'COMPUTER', #2:'SCIENCE'} to encode 'COMPUTER SCIENCE COMPUTER'.

### Two dictionary entries

Predict the first step from the prompt.

### #1 #2 #1

Explain the intermediate result or decision.

### Restore original phrase

Treat spaces as retained separators: #1 #2 #1. Decoder substitutes entries to recover 'COMPUTER SCIENCE COMPUTER'. Include dictionary storage when calculating total size.

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Explain dictionary compression and why the decoder needs the dictionary. [6]

### Question 2

Using entries {1:'RED',2:'BLUE'}, encode 'RED BLUE RED RED' and decode '2 1'. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Evaluate dictionary coding for a document with frequent terms versus already-compressed random-looking data. [8]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Dictionary coding requires permanently discarding repeated words.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.